

Machine Design Final Project

Brian Baudoin, Milain Marcel, Gavin Murray

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Introduction

Rotary cutting tools are devices that are heavily utilized and relied upon within the field of dentistry. These devices must be able to make precise cuts while maintaining the safety of the user, as well as the patient who is being operated on. Alongside safety, these tools must also be lightweight to reduce fatigue in the user and maximize performance during these delicate procedures. The purpose of this project is to design a mechanically driven dental rotary cutting system that maximizes the ratio of cutting force to total weight with the end goal of optimizing both performance and usability within dental environments.

In this project, we designed a mechanical handpiece powered by an electric motor operating at up to 30,000 rpm and 2 N*cm of torque. The designed system must fit within given size limits and include a mechanical drivetrain to transmit power from the source to the drill bit. This drivetrain will be made up of gears, shafts, and bearings that all work simultaneously in the process of power transmission, resulting in a capable and reliable cutting device.

This project utilizes iterative mechanical design, modeling through CAD, and analytical stress evaluation. All mechanical components are modeled and assembled in Fusion. The system's critical loads, potential failure points, and operating stresses are determined using hand calculations. These analyses incorporate shaft and gear contact and bending stresses, factors of safety, mass, weight, and verification of failure criteria.

Design Method

Materials and Design Specifications

The design specifications for this project were provided directly in the prompt. From the beginning, we were given a schematic of the handpiece along with the constraints for all the key parameters. In the initial stages of the project, we worked directly from this schematic to get a general idea of how the device would need to be designed. From that early layout, we found that we would need to use bevel gears to drive the input from the motor to the bur cutting surface. This allowed the system to stay compact while still transferring the motion where it needed to go.

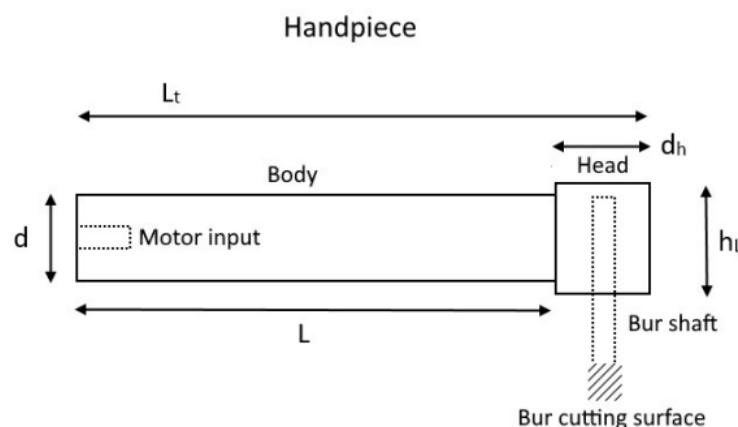


Figure 1. Original schematic of the handpiece

Some dimensional constraints include the size of the head diameter which restricts the size of the bevel gears and bearings. However, this small size is important for maneuverability and visibility inside the patient's mouth. Furthermore, the total length was constrained, impacting the device's overall ergonomics and balance. Some operational constraints include the factor of safety being above 2, ensuring the shafts and gears are safe and durable for long term use in a medical setting, preventing failure. Another is a lightweight feel, as this constrains material selection and component size to reduce user fatigue and maximize precision during procedures. Finally, the materials must be biocompatible as the hand piece cannot cause an adverse reaction with tissues or fluids. With this, the material must be able to be sterilized and withstand repeated autoclaving cycles, which restricts material choices to those with high corrosion and thermal resistance to maintain safety.

Below is a table that includes the geometry specifications and the final values we selected. For several of these parameters, a maximum value was provided, but we intentionally did not use the max options. This was done to maintain a lightweight device and to give the handpiece a more aesthetic and balanced overall form.

Specification	Constraint Value	Chosen Value
Total Handpiece Length = L_t	Max = 130 mm	122 mm
Head diameter = d_h	Max = 22 mm	22 mm
Head height = h_l	Max = 26 mm	26 mm
Handpiece body diameter = d	Max = 28 mm	20 mm
Handpiece body length = L	Max = 112 mm	100 mm
Motor input shaft length = L_m	Max = 25 mm	25 mm
Motor input shaft diameter = d_m	Max = 7 mm	7 mm
Bur shaft diameter = D_b	1.6 mm	1.6 mm
Bur total length = L_b	34 mm	34 mm
Bur cutting surface diameter	2 mm	2 mm
Bur cutting surface length	4 mm	4 mm

Figure 2. Geometry specification table

Based on the initial schematic, we identified the specific components that we would need in order to create our gear train configuration. From the identification we determined that we would have a bevel gear system that was driven by an electric motor to then cut the burr. This is because of the bevel gears ability to transmit power between shafts that are intersecting at the 90 degree angle we wanted to achieve. The design would also include bearings to allow for the shafts to rotate and drive the movement of the gears.

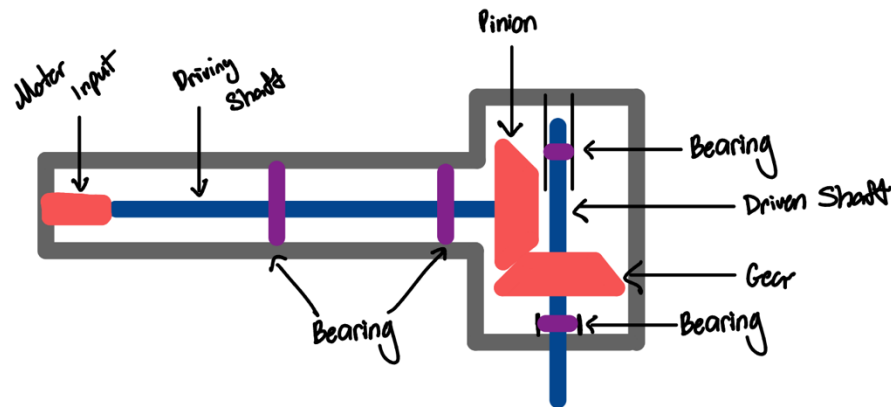


Figure 2. Comparison of provided schematic and design CAD model

This led into the process of material selection for each major component. For the gear and shaft system, AISI 1095 steel was selected because of its high strength for the loads expected from the gear train (AZoM, 2012; QILU Steel Solution, 2024). For the handpiece housing, we used Stainless Steel 316 because of its biocompatibility, corrosion resistance, and its common use in medical devices (Brown, 2025).

For the cutting burr, we wanted to use a material commonly used in the dental industry. After researching industry standards, we found that tungsten carbide is widely used for dental cutting burrs because of its cutting performance and long-term durability (Tungsten Carbide Dental Drills: Uses, Benefits & Comparisons, n.d.).

Finally, for the bearing selection, we looked at common bearing materials and standard practices in similar systems. We chose Chrome Steel based on its strength, durability, and reliability under constant rotational loading (Composites, 2025).

The forces that are involved in the functioning of the cutting system are the input loads from the motor, the internal loads from the gears, and the output loads at the bur. The input load from the motor is the rotational torque delivered by the electric motor to the driving shaft. This is relevant as this is the primary input load used to calculate the mean torsional shear stress on the shafts. The internal loads occur when the bevel gears mesh and the input torque is translated into three forces that act on the shafts and bearings. These forces being the cause of the bending and axial stresses. Finally, the output load is the cutting force, which is the resisting force exerted by the biological material against the tungsten carbide burr as it cuts.

Stress Analysis of Gears

For the gears, we decided to use a 1:1 gear ratio. Choosing this ratio allowed us to simplify the calculations needed to determine the forces acting on the shaft and throughout the gear train. To achieve this, we selected a 32-tooth gear and used the general standard value of a 20° pressure angle. During the design process, we also made several reasonable assumptions and kept the bearing spacing the same on each shaft. This was intentional, as keeping the distances consistent helps simplify the hand calculations and reduces unnecessary complexity in the analysis.

Needed Variables	Symbol	Value	units	Equation
Pitch line velocity	V	6283.185307	ft/min	$V = \pi \cdot d \cdot n / 12$
Transmitted Load	Wt	0.442536	lbf	$Wt = 33000 \cdot H / V$
Overload Factors	Ko	1		Table 15-2
Dynamic Factors	Kv	1.168076208		$Kv = (A + (\text{SQRT}(V)) / A)^B$
	A	92		Eq. 15-6, $A = 50 + 56(1-B)$
	B	0.25		Eq. 15-6, $B = 0.25(12 - Qv)^{(2/3)}$
Size factor	Ks	0.5		15-10, $P_d > 16$ teeth/in
Kmb- needed for Km	Kmb	1.1		one member straddle mounted
Load distribution factor	Km	1.100225		$Km = Kmb + 0.0036 \cdot F^2$
Lengthwise curvature factor	Kx	1		15-13
Geometry factor for Pinion	(J)P	0.215		Fig. 15-7
Geometry Factor for Gear	(J)G	0.215		Fig. 15-7
Elastic Coefficient	Cp	2290	psi ^{0.5}	15-21, assume steel
Geometry factor for pitting resistance	I	0.065		Fig. 15-6
Size factor for pitting resistance	Cs	0.46875		$Cs = 0.125F + 0.4375$
Crowning factor for pitting resistance	Cxc	1.5		15-12
Allowable bending stress number	Sat	13012	psi	$Sat = 44(HB) + 2100$ psi
Pinion Stress Cycle Factor for bending	(KL)P	0.9999634264		$1.683NL^{-0.0323}$
Gear Stress Cycle Factor for bending	(KL)G	0.9999634264		$1.683(NL/mG)^{-0.0323}$
Temperature Factor	KT	1		15-18, assume $t < 250$ degrees
Reliability Factor	KR	1		$0.50 - 0.25 \log(1-R)$
Allowable contact stress number	Sac	108188	psi	$Sac = 341(HB) + 23620$ psi
Pinion Stress-Cycle Factor for pitting	(CL)P	1.319634652		$3.4822NL^{-0.0602}$
Gear Stress-Cycle Factor for pitting	(CL)G	1.319634652		$3.4822(NL/mG)^{-0.0602}$
Hardness-Ratio Factor	CH	1		$CH = 1 + B1(mG - 1)$
	B1	0.00069		$B1 = 0.00898 (HBP/HBG) - 0.00829$
Contact Safety Factor	SH	2		
Reliability Factor	CR	1		$CR = \text{SQRT}(KR)$

Figure 3. Relevant variables and values for gear analysis

Once those base values were confirmed, we calculated both the bending stress and the contact stress for the pinion and the gear. After running these calculations, we verified that the selected gears would operate safely under the expected load. The factor of safety for contact stress was found to be 3.4, and the factor of safety for bending stress was 7.8. These values were the same for both gears and confirmed that the design met the necessary strength requirements.

Stress Analysis				
Variable	Symbol	Value	Units	Equation
Pinion bending				
Bending stress	(St)P	211.6181045	psi	$(St)P = (Wt/F) \cdot Pd \cdot Ko \cdot Kv \cdot ((Ks \cdot Km) / (Kx \cdot JP))$
Allowable bending stress	(Sw)P	6505.762052	psi	$(Sw)P = (Sat \cdot (KL)P) / (SF \cdot KT \cdot KR)$
Bending factor of safety for the pinion	(SF)P	7.841292876		$(SF)P = ((Sat \cdot (KL)P) / (KT \cdot KR \cdot (St)P))^{0.5}$
Gear bending				
Bending stress	(St)G	211.6181045	psi	$(St)G = (Wt/F) \cdot Pd \cdot Ko \cdot Kv \cdot ((Ks \cdot Km) / (Kx \cdot JG))$
Allowable bending stress	(Sw)G	6505.762052	psi	$(Sw)G = (Sat \cdot (KL)G) / (SF \cdot KT \cdot KR)$
Bending factor of safety for the gear	(SF)G	7.841292876		$(SF)G = ((Sat \cdot (KL)G) / (KT \cdot KR \cdot (St)G))^{0.5}$
Pinion wear				
Contact stress	(Sc)P	12700.78963	psi	$(Sc)P = Cp \cdot ((Wt/F \cdot dp) \cdot Ko \cdot Kv \cdot Km \cdot Cs \cdot Cxc)^{0.5}$
Allowable contact stress	(Swc)P	71384.31688	psi	$(Swc)P = (Sac \cdot (CL)P \cdot CH) / (SH \cdot KT \cdot CR)$
Wear factor of safety for the pinion	(SH)P	3.352748991		$(SH)P = ((Sac \cdot (CL)P \cdot CH) / (KT \cdot CR \cdot (Sc)P))^{0.5}$
Gear wear				
Contact stress	(Sc)G	12700.78963	psi	$(Sc)G = Cp \cdot ((Wt/F \cdot dp) \cdot Ko \cdot Kv \cdot Km \cdot Cs \cdot Cxc)^{0.5}$
Allowable contact stress	(Swc)G	71384.31688	psi	$(Swc)G = (Sac \cdot (CL)G \cdot CH) / (SH \cdot KT \cdot CR)$
Wear factor of safety for the gear	(SH)G	3.352748991		$(SH)G = ((Sac \cdot (CL)G \cdot CH) / (KT \cdot CR \cdot (Sc)G))^{0.5}$

Figure 4. Calculations of gear stress and factors of safety

Shaft Stress Analysis

For the shaft stress analysis, we initially found the forces transmitted by the bevel gears, using the known motor torque (T) and gear geometry we calculated the tangential force, radial force, and axial force. The resultant force was then found for the driving and driven shaft by adding the moments from each plane using the Pythagorean theorem (Figure B.1). We then used superposition in order to determine the resulting moment acting on each of the shafts. This approach allowed us to break the loading into simpler components and then add them together to get the total moment, which was necessary for evaluating the stresses in the shaft.

The alternating moment was found to be equal to the absolute value of the maximum moment and the mean torsion is equal to the torsion of the gear, while the alternating torsion and mean moment are equal to 0 (Figure B5). This is because the torsional load is constant caused by the motor's output torque which would be operating steadily at a set speed. Furthermore, because there is constant torsion and completely reversed bending stress cycle due to the stress cycles are equal and opposite maximum values due to the 360 degree rotation, the alternating torsion and mean moment are equal to 0. From here, because our shafts were unnotched the stress concentration factors of k_f and k_{fs} were set to 1 because the stress distribution is uniform across the cylindrical shaft. The alternating and mean normal and shear stresses were then calculated (Figure B7 & B8) with mean normal stress and alternating shear being 0. Because the shaft operated under a multiaxial stress state it has both bending and shear stress acting simultaneously, so we decided to then use the Von Mises stress equations to find the multiaxial stress state at a point into a single, equivalent uniaxial stress. The alternating and mean Von Mises were then found to find the total stress amplitude that is driving the fatigue damage and the total effective average stress that reduces the material's fatigue strength (Figure B9).

We then applied the DE-Goodman Criterion to determine whether the shaft would fail under the combined loading conditions. To use this criterion, we had to calculate both the alternating stress and the mean stress and then compare these values against the material limits. As part of this process, we used Marin's equation along with the standard correction factors to solve for the modified endurance limit, S_e , and the ultimate tensile strength, S_{ut} . Since our shafts were unnotched, we assumed that many of the k values were effectively equal to 1, which simplified the analysis while still giving conservative results.

$$\begin{array}{l}
 \underline{S_{ut} \text{ and } S'_e} \\
 * \text{ Table A-20} \\
 \text{AISI 1045 HR} \\
 S_{ut} = 120 \cdot 10^3 \text{ psi} \\
 S_{ut} \leq 200 \cdot 10^3 \text{ psi} \\
 \rightarrow S'_e = 0.5 S_{ut} \\
 0.5(120 \cdot 10^3) = 60 \cdot 10^3 \text{ psi}
 \end{array}$$

Figure 5. Calculations of ultimate tensile strength and endurance limit

$$\begin{array}{l}
 \underline{\text{Correction Factors}} \\
 K_a = a S_{ut}^b \quad S_{ut} = 120 \text{ kpsi} \\
 \text{For HR} \rightarrow a = 11 \text{ kpsi} \\
 \text{(Table 6-2)} \quad b = -0.650 \\
 K_a = 11 (120)^{-0.650} \\
 = 0.4890 \\
 K_b = 1 \text{ (assumed for unnotched shaft)} \\
 K_c = 1 \text{ (assumed for shaft in torsion and other loads)} \\
 K_d = 1 \text{ (assumed for } S_e \text{ from } S_{ut} \text{ at room temp.)} \\
 K_e = 0.968 \text{ (from Table 6-4 for reliability of 95\%)}
 \end{array}$$

Figure 6. Correction Factor Calculations and Assumptions

Solving for S_e with Marin's equation

$$S_e = K_a(K_b)(K_c)(K_d)(K_e)(S'_e)$$

$$= 0.4890(1)(1)(1)(0.868)(60)$$

$$S_e = 25.47 \text{ kpsi} = 25.47 \cdot 10^3 \text{ psi}$$

Verify DE Goodman Criteria

$$\frac{\sigma'_a}{S_e} + \frac{\sigma'_m}{S_{ut}} \geq \frac{1}{n} ; n \geq 2$$

* Driving Shaft

$$\frac{111.2}{(25.47 \cdot 10^3)} + \frac{74.6}{(120 \cdot 10^3)} \geq \frac{1}{2}$$

$$0.005 \geq 0.5$$

Shaft does not fail

* Driven Shaft

$$\frac{1640.95}{(25.47 \cdot 10^3)} + \frac{6247.13}{(120 \cdot 10^3)} \geq \frac{1}{2}$$

$$0.1165 \geq 0.5$$

Shaft does not fail

Figure 7. Calculations of S_e and verification of DE Goodman criteria

Design Changes

Our initial design was followed fairly closely, but there were a few changes made as we continued to work through the calculations and modeling. Some of the key adjustments included relocating the bearings and modifying the thickness of the driving shaft. These updates were primarily made to better align the design with the mechanical requirements and to simplify the analysis as we progressed.

One of the main modifications was increasing the thickness of the driving shaft. This change was necessary to account for the coupling required to connect the shaft to the motor input, which had a diameter of 25 mm. To address this, we planned to use a muff coupling, which allows the two rods of matching dimensions to overlap and then be secured together. Since this type of coupling is a rigid coupling, it would help ensure that the components remain secure with minimal tolerance for movement (Admin, 2017).

Another change involved the placement of the bearings. The approximate bearing locations were originally estimated to be at the beginning and midpoint of the driving shaft. However, we later adjusted this so that the distance between the bearings matched the distance between the bearing and the gear. This change made the hand calculations more straightforward and created a more consistent spacing that made it easier to iterate the design.

Overall Design Weight

The overall design ended with a weight of 0.45lbs once all the components are assembled inside. This calculation took into account the varying densities of the materials and the sizes of each of the components. The final design met the goals of having a light weight device as well as maintaining a factor of safety of at least 2. The drill composition could continue to be altered to make it lighter through small changes in the model such as decreasing the height of the gears as well as the overall thickness of the shafts.

Summary and Conclusions

The final overall design of the dental cutting tool successfully meets project requirements and constraints. Safety, durability, and performance are all thoroughly analyzed and verified through various calculations and criteria. Through the process of iterative modeling and analysis, the system is able to effectively transmit power from the 30000 rpm source to the cutting burr while being characterized as lightweight and sturdy.

Future manufacturing of the design would likely involve multiple individual processes to mass produce this tool. Machine processes such as milling and the use of a lathe would be required for the creation of the shafts and housing. Precise cutting and grinding would be needed for production of the gear system along with some form of press device for the fitting of the bearings. The choice to use Stainless Steel 316 for the housing and AISI 1095 (carbon steel) for the gears and shafts to ensure a strong, durable, and biocompatible final product. Choosing tungsten carbide for the burr further builds upon the significance of biocompatibility and strength in the tool.

The various engineering calculations performed throughout this project indicate that the drivetrain will operate efficiently and safely within a dental procedure environment. The gears achieved a factor of safety of 3.4 for contact stress and 7.8 for bending stress which confirms they will withstand the transmitted torque with no risk of failure. The DE Goodman Criteria used for shaft analysis also verifies that the endurance limit exceeds both alternating and mean stresses which further builds on the safety of the design. Finally, the final mass of the tool is approximately 0.45 lbs. and satisfies the goal of maintaining a lightweight handpiece suitable for prolonged use during dental procedures. This is crucial for the working environment as it minimizes fatigue and reduces tremor in the user's hand during prolonged procedures, ensuring precision. Furthermore, the high factors of safety with the gears being above 3 and the shafts being way greater than 1, are well prepared for the expected loads. This high reliability is important in a clinical setting, guaranteeing the device will not fail during a procedure.

Overall, the results show that the design is realistic, manufacturable, and capable as required in such a field of work. While the system meets the necessary criteria, future improvements may consist of further improving comfortability while being held and vibration reduction. Even before such improvements, this design creates a strong and supported foundation for an efficient and reliable dental rotary cutting tool.

References

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Appendix

Appendix A. Designed Components

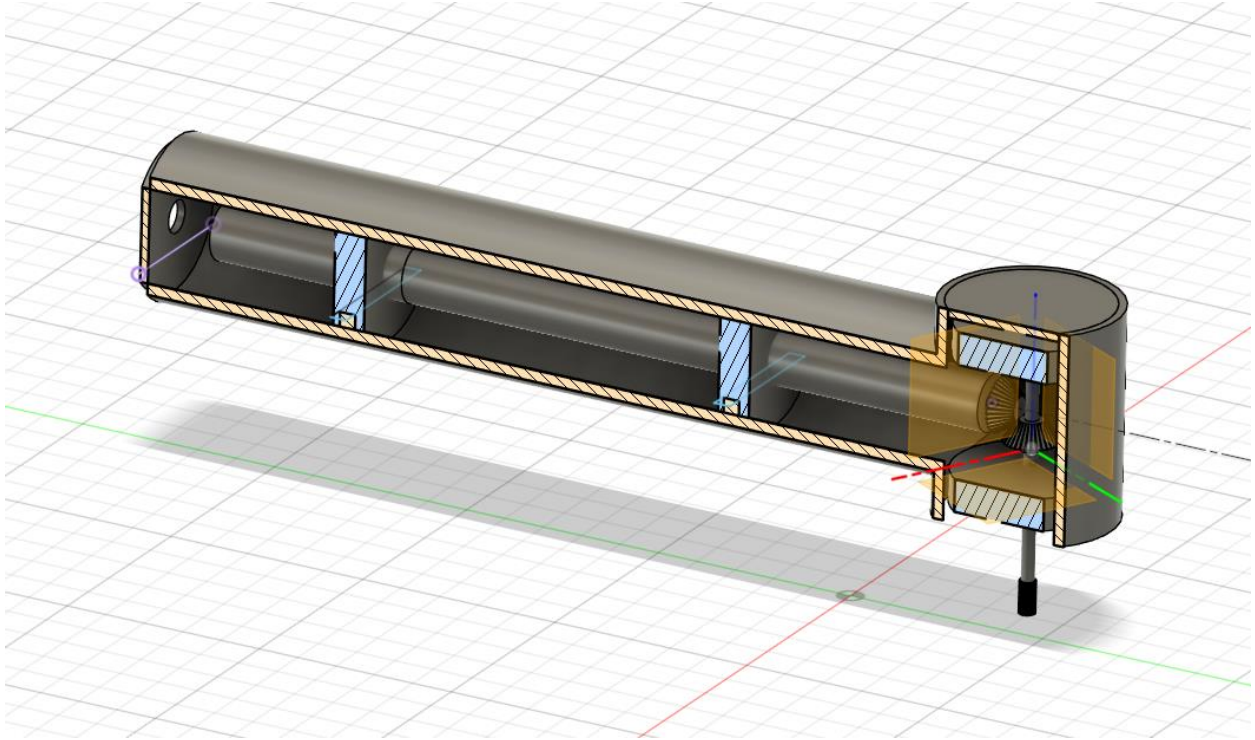


Figure A1. Overall cross-section of design with drivetrain view

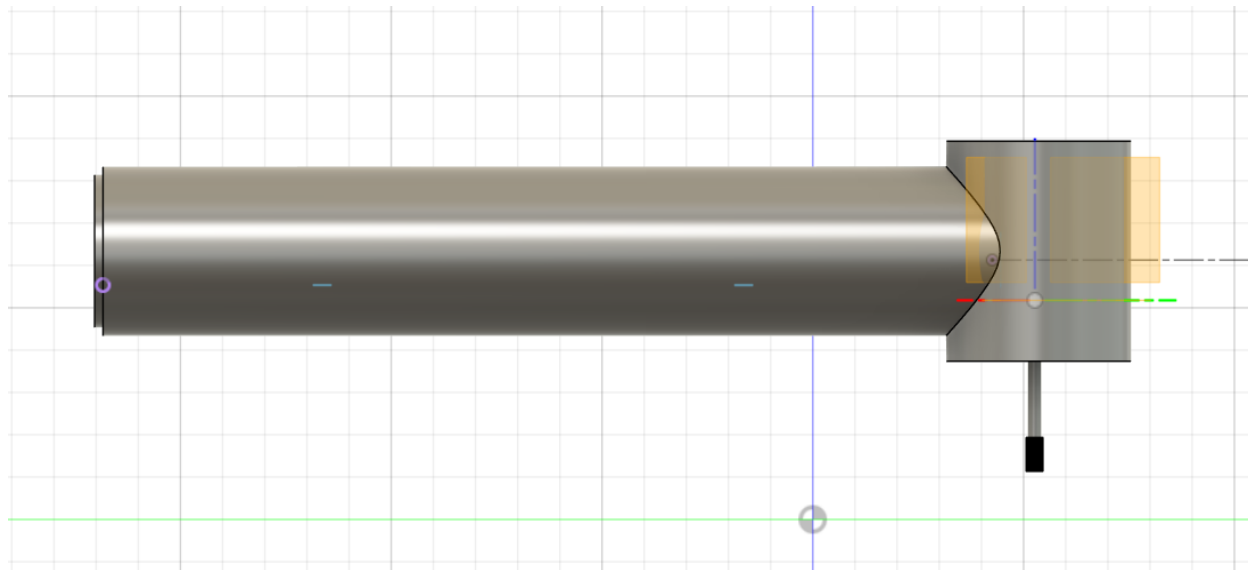


Figure A2. Front view of assembly

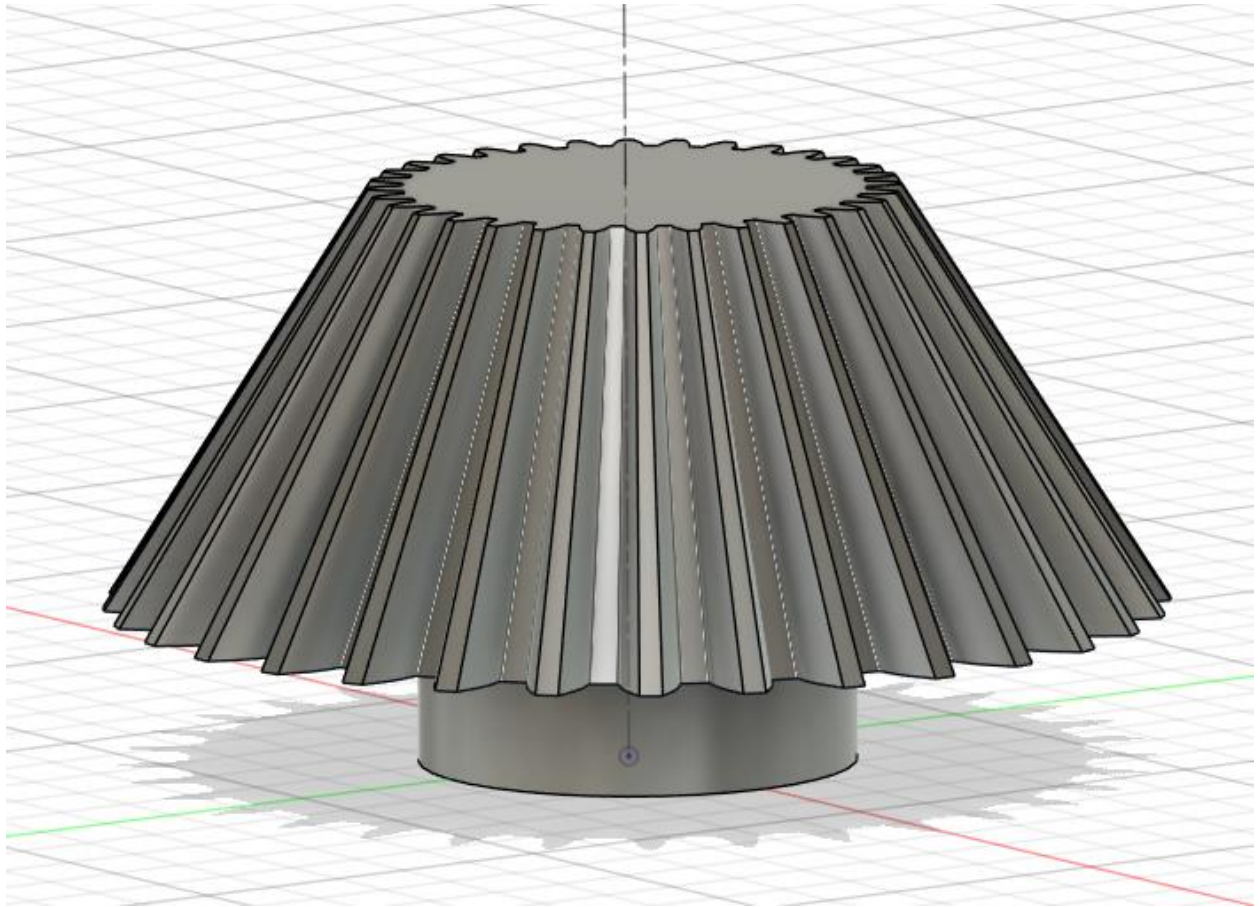


Figure A3. Front view of bevel gear

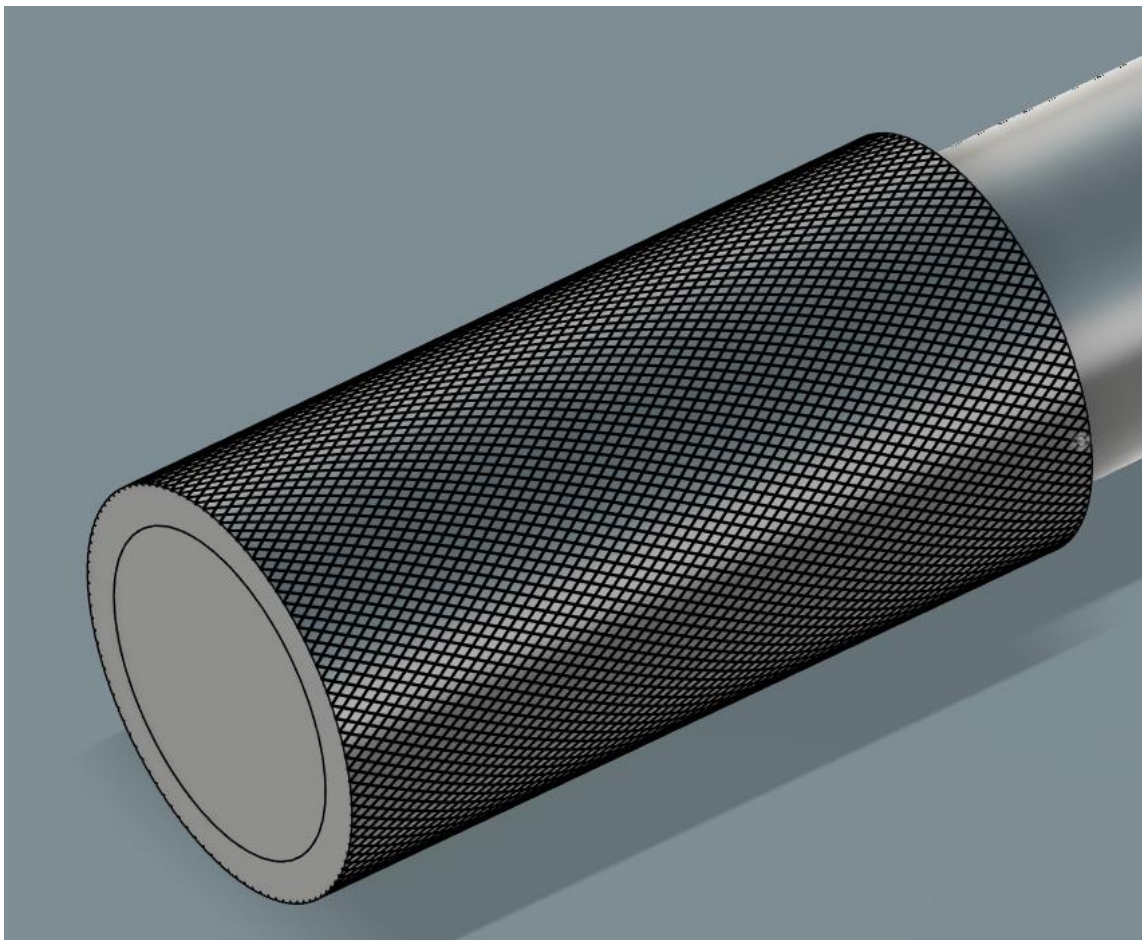


Figure A4. Driving interface of driven shaft



Figure A5. Overall view of driven shaft

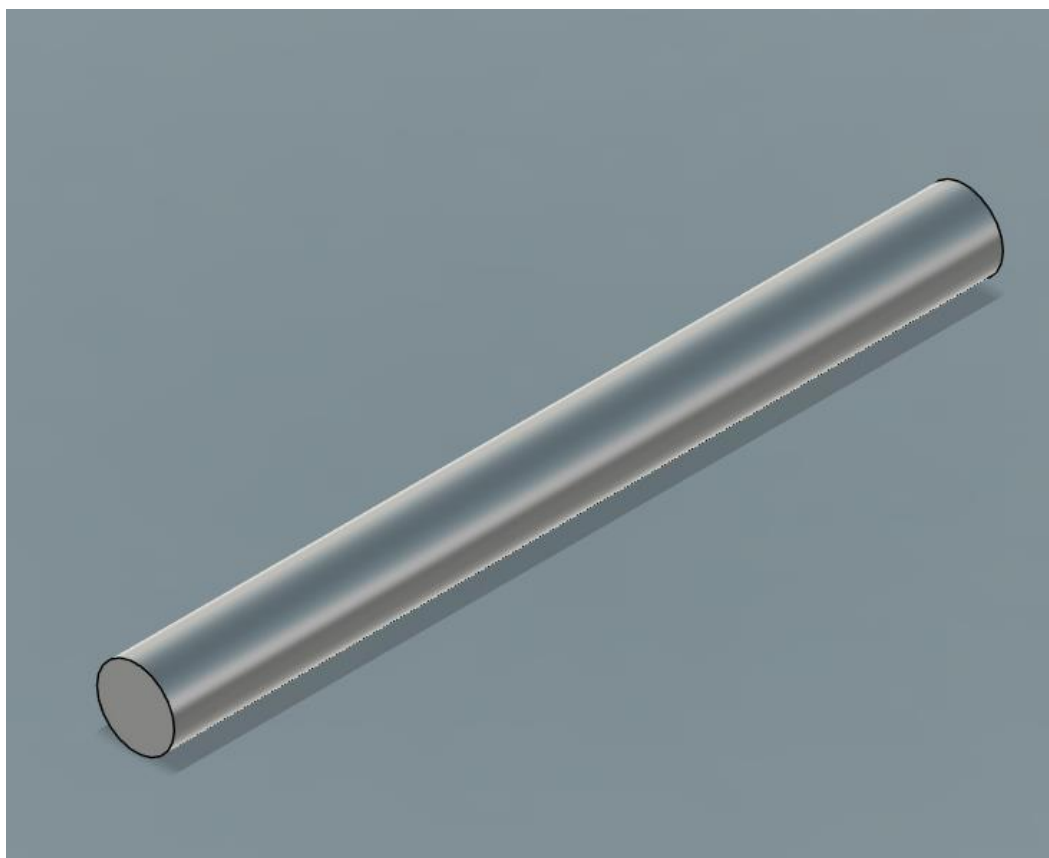


Figure A6. Overall view of driving shaft

Appendix B. Engineering Calculations

Finding Gear Forces

$$W_r = 0.442536 \text{ lbf}$$

$$\phi = 20^\circ$$

$$\Gamma = 45$$

$$W_r = W_t (\tan \phi) \cos \Gamma = 0.1139 \text{ lbf}$$

$$W_a = W_t (\tan \phi) \sin \Gamma = 0.1139 \text{ lbf}$$

Driving Shaft

$$\begin{aligned} F_1 &= \sqrt{W_t^2 + W_r^2} \\ &= \sqrt{0.44^2 + 0.1139^2} \\ &= 0.457 \text{ lbf} \end{aligned}$$

Driven Shaft

$$\begin{aligned} F_2 &= \sqrt{W_r^2 + W_a^2} \\ &= \sqrt{0.1139^2 + 0.1139^2} \\ &= 0.161 \text{ lbf} \end{aligned}$$

Max Max

Driving

$$\begin{aligned} M_{\max} &= -F_1 (LAP) \\ &= -0.4569 \cdot (0.5) \\ &= -0.228 \text{ lbf} \cdot \text{in} \end{aligned}$$

$$\begin{aligned} M_a &= 0.228 \\ m_m &= 0 \\ T_a &= 0 \\ T_m &= 0.01475 \end{aligned}$$

Driven

$$\begin{aligned} M_{\max} &= \frac{F_2 (L_{cd}) (L_{cb})}{(L_{cd} + L_{cb})} \\ &= \frac{0.161 (1.969) (0.5)}{(0.5 + 0.5)} \\ &= 0.040267 \end{aligned}$$

$$\begin{aligned} M_a &= 0.040267 \\ m_m &= 0 \\ T_a &= 0 \\ T_m &= 0.01475 \end{aligned}$$

Figure B1. Shaft calculations pt. 1

S_{ut} and S'_e

* Table A-20

AISI 1095 HR

$$S_{ut} = 120 \cdot 10^3 \text{ psi}$$

$$S_{ut} \leq 200 \cdot 10^3 \text{ psi}$$

$$\rightarrow S'_e = 0.5 S_{ut}$$

$$0.5(120 \cdot 10^3) = 60 \cdot 10^3 \text{ psi}$$

Figure B2. Ultimate tensile strength and uncorrected endurance limit calculations

Correction Factors

$$K_a = a S_{ut}^b \quad S_{ut} = 120 \text{ kpsi}$$

$$\begin{aligned} \text{For HR} &\rightarrow a = 11 \text{ kpsi} \\ (\text{Table 6-2}) &\quad b = -0.650 \end{aligned}$$

$$\begin{aligned} K_a &= 11 (120)^{-0.650} \\ &= 0.4890 \end{aligned}$$

$$K_b = 1 \quad (\text{assumed for unnotched shaft})$$

$$K_c = 1 \quad (\text{assumed for shaft in torsion and other loads})$$

$$K_d = 1 \quad (\text{assumed for } S_e \text{ from } S_{ut} \text{ at room temp.})$$

$$K_e = 0.868 \quad (\text{from Table 6-4 for reliability of 95\%})$$

Figure B3. Correction factor calculations

Solving for S_e with Marin's equation

$$S_e = K_a(K_b)(K_c)(K_d)(K_e)(S'_e) \\ = 0.4890(1)(1)(1)(0.868)(60)$$

$$S_e = 25.47 \text{ kpsi} = 25.47 \cdot 10^3 \text{ psi}$$

Verify DE Goodman Criteria

$$\frac{\sigma'_a}{S_e} + \frac{\sigma'_m}{S_{ut}} \geq \frac{1}{n} ; n \geq 2$$

* Driving Shaft

$$\frac{111.2}{(25.47 \cdot 10^3)} + \frac{74.6}{(120 \cdot 10^3)} \geq \frac{1}{2}$$

$$0.005 \geq 0.5$$

Shaft does not fail

* Driven Shaft

$$\frac{1640.95}{(25.47 \cdot 10^3)} + \frac{6247.13}{(120 \cdot 10^3)} \geq \frac{1}{2}$$

$$0.1165 \geq 0.5$$

Shaft does not fail

Figure B4. Calculation of corrected endurance limit and failure criteria

Driving shaft

$$M_{max} = -F_1(LAP)$$

$$= 0.45696(0.5)$$

$$= -0.22848 \text{ lbf}\cdot\text{in}$$

Driven shaft

$$M_{max} = \frac{F_2(L_{CG})(L_{DG})}{L_{CG} + L_{DG}}$$

$$= \frac{0.161(0.5)(0.5)}{0.5 + 0.5}$$

$$= 0.0403 \text{ lbf}\cdot\text{in}$$

Figure B5. Shaft calculations pt. 2

$$M_A = \left| \frac{M_{max} - (-M_{max})}{2} \right| = M_{max} \quad T_m = T_{gear}$$

$$M_m = 0 \quad \text{b/c completely reversed bending}$$

$$T_d = 0 \quad \text{and constant torsion}$$

Driving shaft

$$M_d = 0.22848 \text{ lbf}\cdot\text{in}$$

$$M_m = 0$$

$$T_d = 0$$

$$T_m = 0.01475 \text{ ft}\cdot\text{lbf}$$

$$= 0.177 \text{ lbf}\cdot\text{in}$$

Driven shaft

$$M_d = 0.0403 \text{ lbf}\cdot\text{in}$$

$$M_m = 0$$

$$T_d = 0$$

$$T_m = 0.01475 \text{ ft}\cdot\text{lbf}$$

$$= 0.177 \text{ lbf}\cdot\text{in}$$

Figure B6. Shaft calculations pt. 3

$$\sigma_a = \frac{\sigma_c (32) (M_o)}{\pi d_s^3} \quad \sigma_m = \frac{k_f (32) m_m}{\pi d_s^3}$$

$$\tau_o = \frac{k_{fs} (16) T_o}{\pi d_s^3} \quad \tau_m = \frac{k_{fs} (16) T_m}{\pi d_s^3}$$

B/c Unnotched shafts

$$k_f = 1 \quad k_{fs} = 1$$

Shaft Diameters

$$d_{s1} = 7 \cdot \frac{1}{25.4} \text{ in} = 0.2755 \text{ in}$$

$$d_{s2} = 1.6 \cdot \frac{1}{25.4} = 0.0629 \text{ in}$$

Figure B6. Shaft calculations pt. 4

Driving

$$\sigma_a = \frac{(1)(32)(0.2285)}{\pi (0.2755)^3} \quad \sigma_m = \frac{1(32)(0)}{\pi (0.2755)^3}$$

$$= 111.19 \text{ lbf} \cdot \text{in}^2 \quad = 0$$

$$\tau_o = \frac{1(16)(0)}{\pi (0.2755)^3} \quad \tau_m = \frac{1(16)(0.177)}{\pi (0.2755)^3}$$

$$= 0 \quad = 43.07 \text{ lbf} \cdot \text{in}^2$$

Figure B7. Shaft calculations pt. 5

Driven

$$\sigma_a = \frac{1(32)(0.0403)}{\pi (0.0629)^3}$$

$$= 1640.95 \text{ lbf} \cdot \text{in}^2$$

$$\sigma_m = \frac{1(32)(0)}{\pi (0.0629)^3}$$

$$= 0$$

$$\tau_a = \frac{1(16)(0)}{\pi (0.0629)^3}$$

$$= 0$$

$$\tau_m = \frac{1(16)(0.177)}{\pi (0.0629)^3}$$

$$= 3606.78 \text{ psi}$$

Figure B8. Shaft calculations pt. 6

$$\sigma'_a = \sqrt{\sigma_a^2 + 3\tau_a^2}$$

$$\sigma'_m = \sqrt{\sigma_m^2 + 3\tau_m^2}$$

Driving shaft

$$\sigma'_a = \sqrt{(111.19)^2 + 3(0)^2}$$

$$= 111.19 \text{ psi}$$

$$\sigma'_m = \sqrt{0^2 + 3(43.03)^2}$$

$$= 74.60 \text{ psi}$$

Driven shaft

$$\sigma'_a = \sqrt{1640.95^2 + 3(0)^2}$$

$$= 1640.95 \text{ psi}$$

$$\sigma'_m = \sqrt{0^2 + 3(3606.78)^2}$$

$$= 6247.13 \text{ psi}$$

Figure B9. Shaft calculations pt. 7

Actual Factor of Safety for Shafts

$$n = \frac{1}{\frac{\sigma'_a}{S_e} + \frac{\sigma'_m}{S_{ut}}}$$

Driving Shaft

$$\frac{1}{\frac{161.2}{(25.47 \cdot 10^3)} + \frac{74.6}{(120 \cdot 10^3)}} = n$$

$$n = 200.5$$

Driven Shaft

$$\frac{1}{\frac{1640.95}{(25.47 \cdot 10^3)} + \frac{6247.13}{(120 \cdot 10^3)}} = n$$

$$n = 8.6$$

Figure B10. Calculations of actual factors of safety for shafts

Appendix C: Orthographic and Isometric Drawings

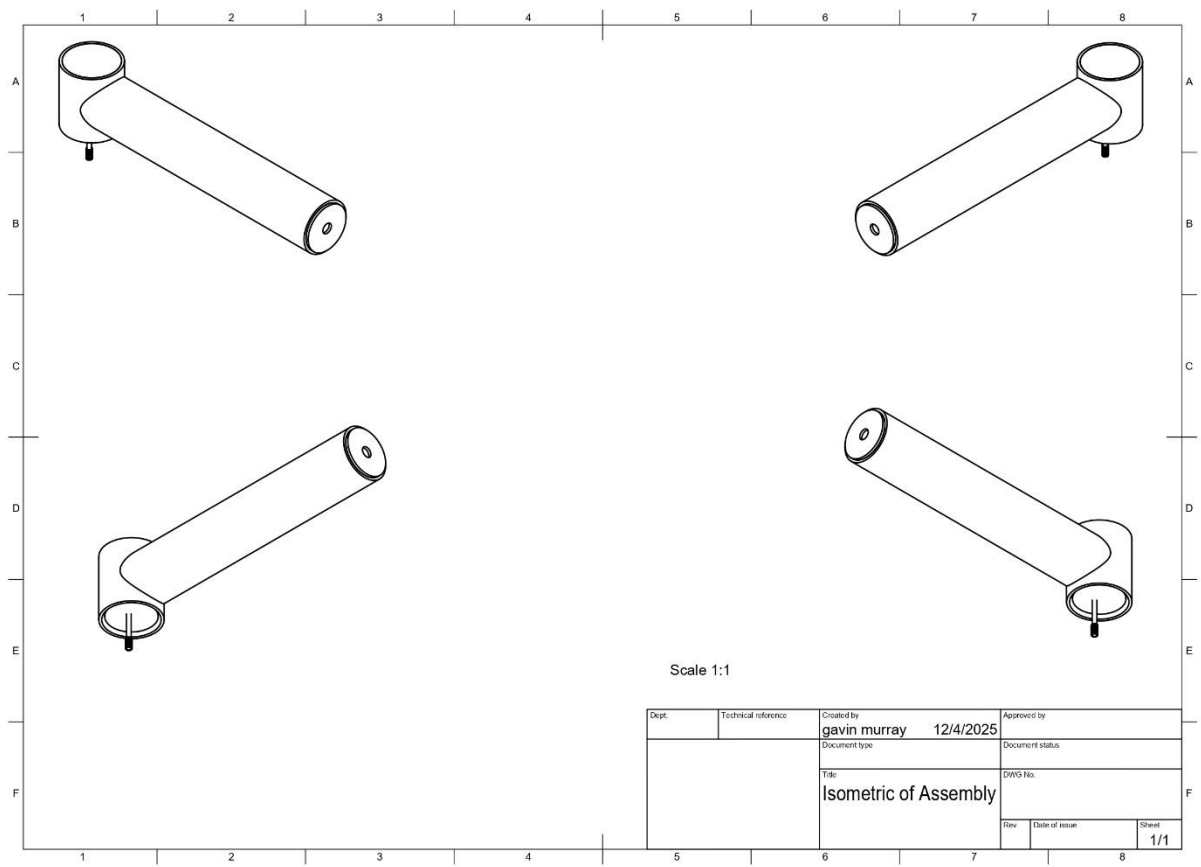


Figure C1. Isometric Drawing of Hand Piece Assembly

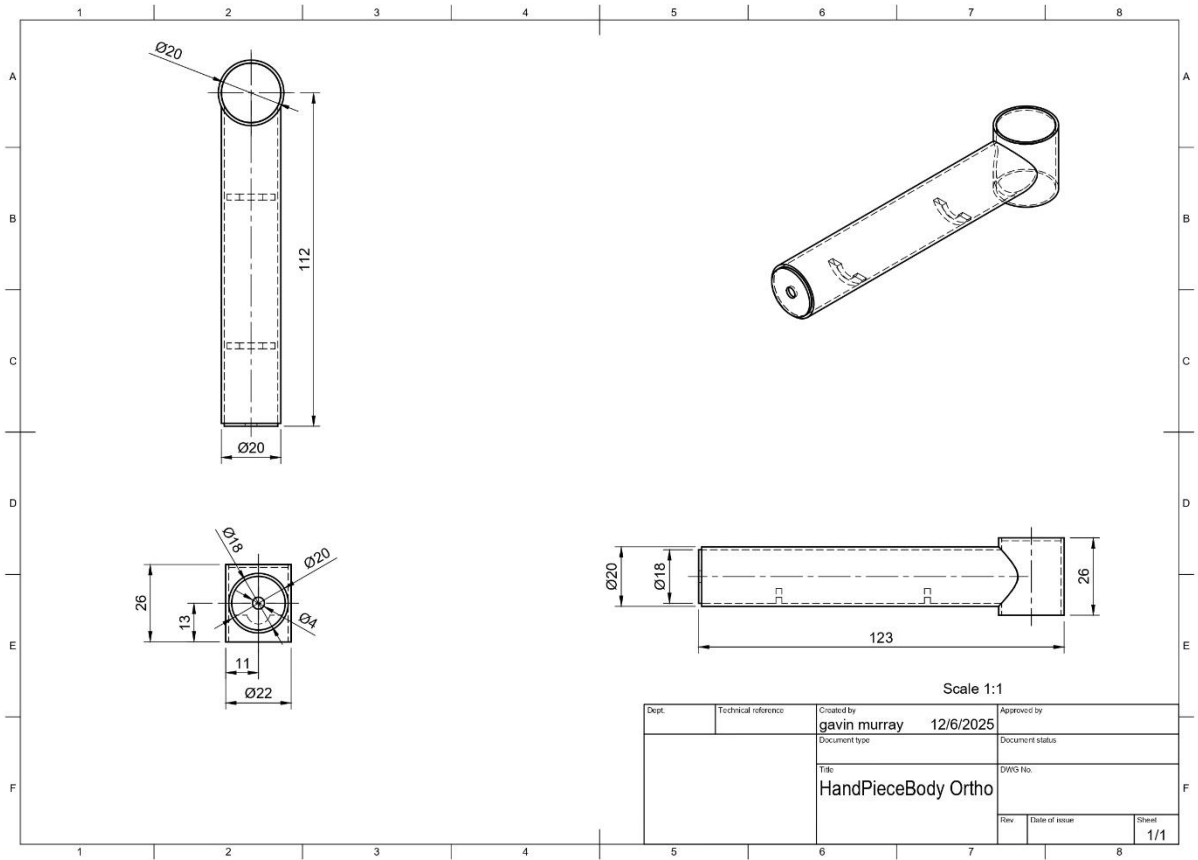


Figure C2. Orthographic Drawing of Hand Piece Body

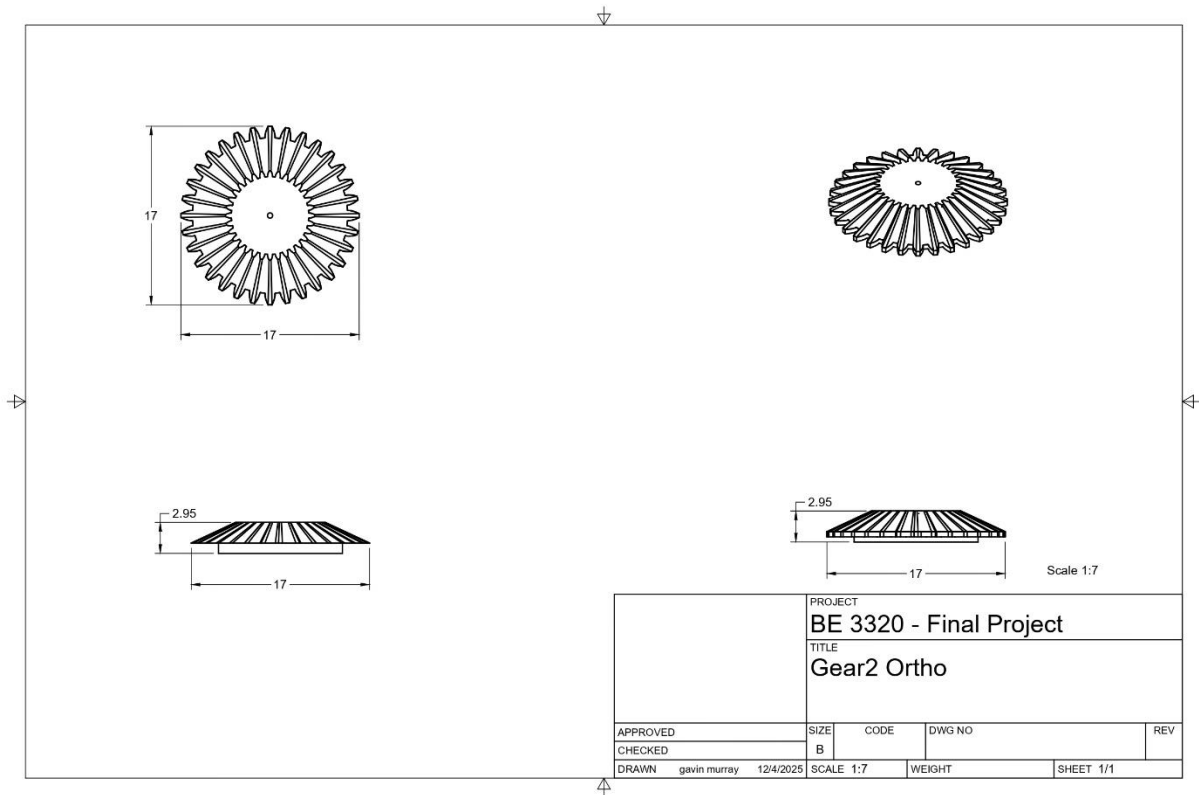


Figure C3. Orthographic Drawing of Gear Design

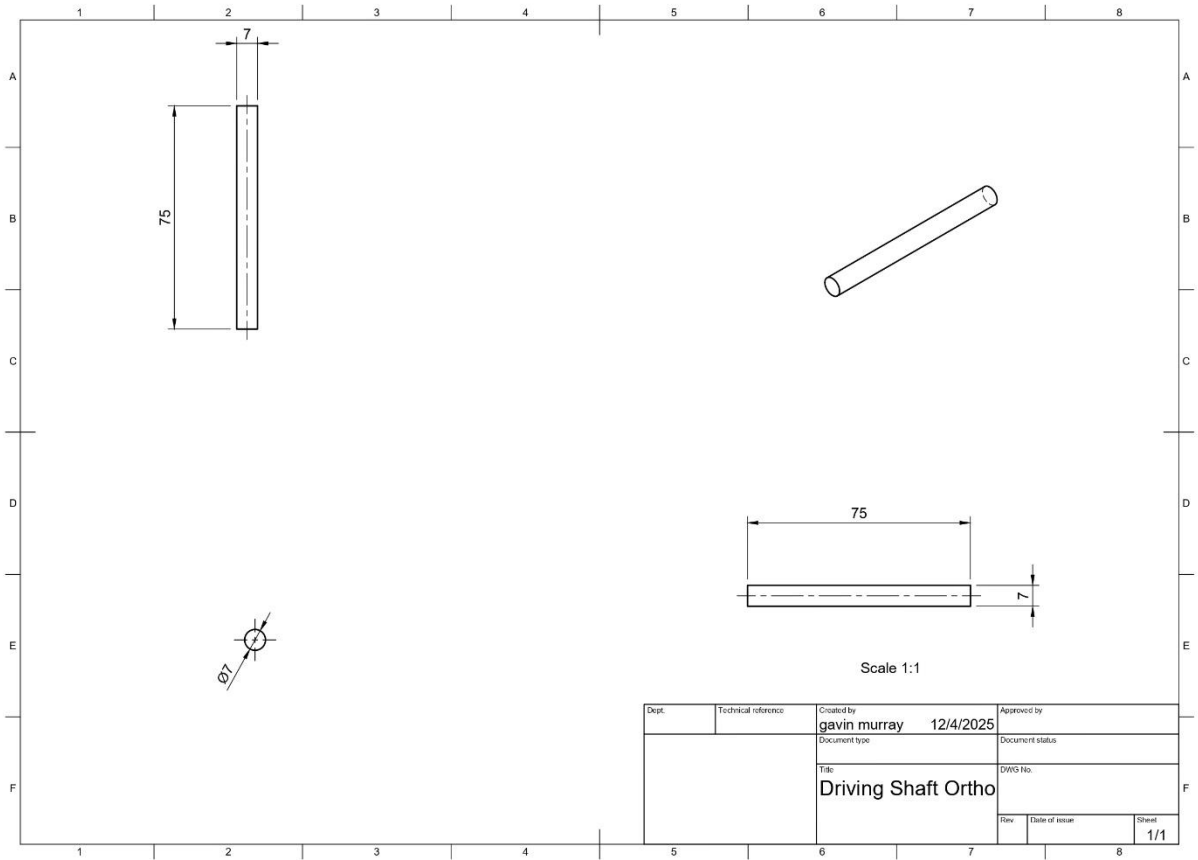


Figure C4. Orthographic Drawing of Driving Shaft

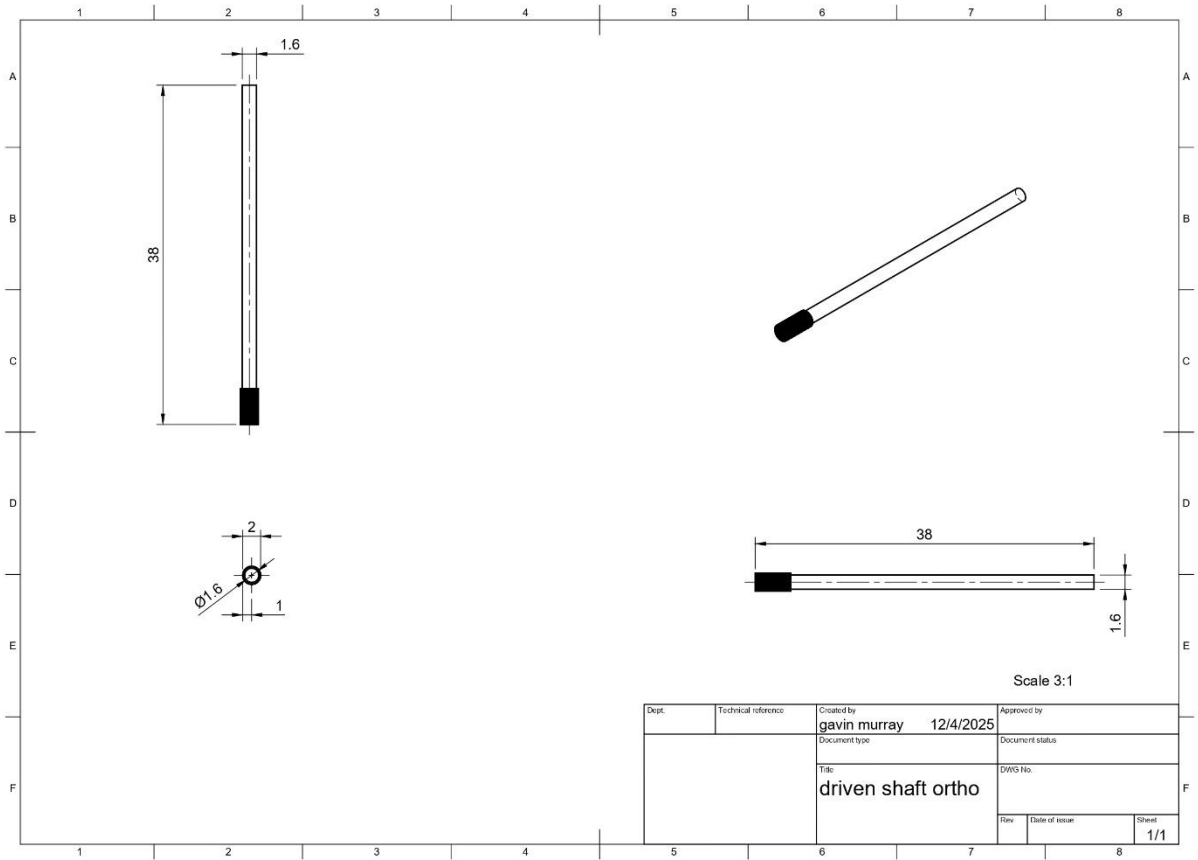


Figure C5. Orthographic Drawing of Driven Shaft

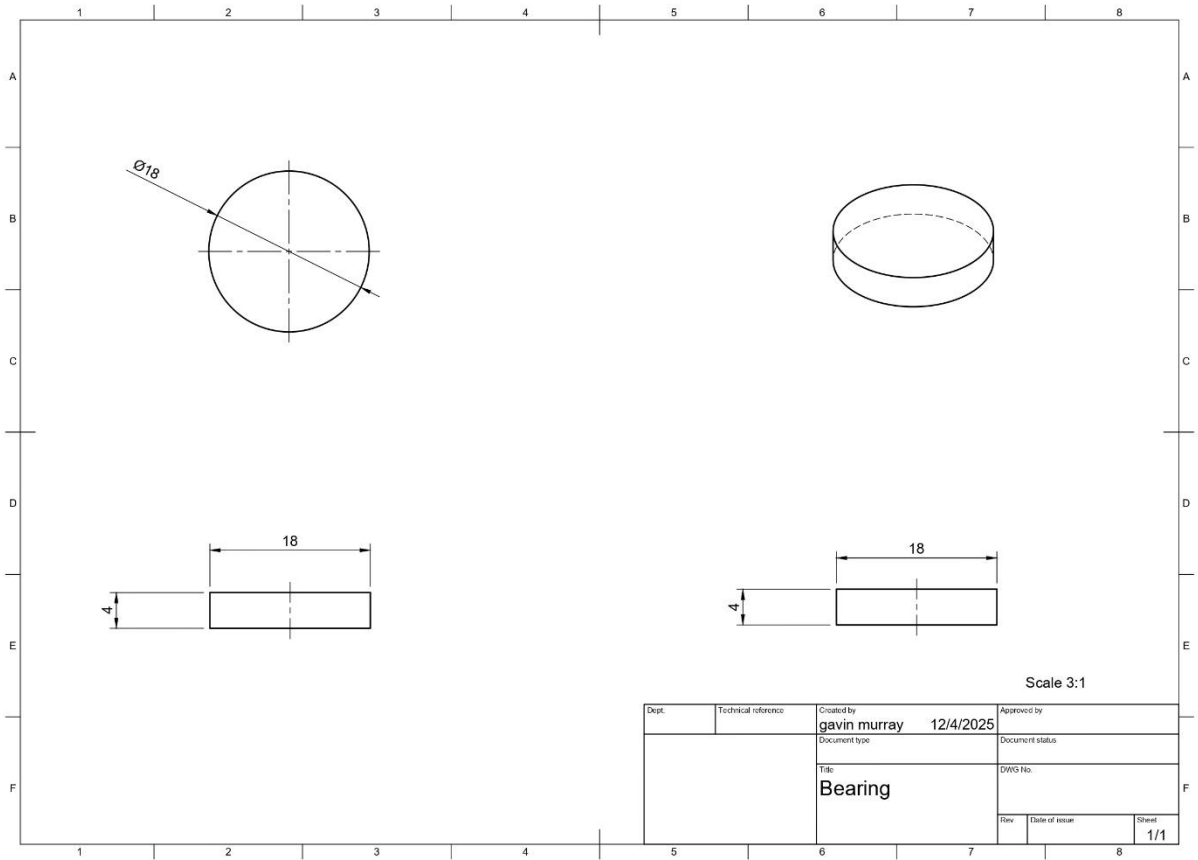


Figure C6. Orthographic Drawing of Bearing Placeholder